Jmar L. Pineda

Electrical and Computer Engineer

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PROFILE

I am currently a freshman studying Electrical And Computer Engineering at NYIT with experience in software development and robotics from working on various personal projects. I have excellent communication and leadership skills that will help companies form a more effective, productive, and fun team environment.

ORGANIZATIONS

Code Coach
theCoderSchool @ Syosset

- Guided children whose ages range from 9-16 y.o. into making their first computer games to teach foundational CS concepts.
- Tracked the learning progress of more than 15 students and gave specialized lessons for each, whether it be Unity, Scratch, Roblox, Arduino, Python, etc.

Embedded Systems Programmer For Robots
QHSS Robotics Club

- Coordinated with the builders and designers of our club to figure out the intended movements and goals for the robot.
- Made sure that our robot was ready and optimized to compete against other schools in competitions.

Programming Tutor
QHSS Hack Club

- Taught programming to my classmates who were interested in learning how to make games and websites.
- Gave them reasonable challenges to complete and gave assistance whenever they were stuck on a problem.

Hackathon Leader
QHSS Hack Club

- Led my team to 1st place during our school-hosted Hackathon by designating tasks and coordinating our efforts while I tackled the hardest problems.

Christian Member and Volunteer
Members Church of God International

- Lifelong christian and volunteer at various church-hosted community service opportunities such as blood drives, food drives, clean-up drives, fundraisers, and setting up free grocery stores.

SKILLS

- Communication
- Leadership
- Teamwork
- Game development with Unity, Godot, and Roblox
- AI Implementations
- Forex/Stock trading algorithm development
- Web development
- Native proficiency in English and Filipino
- Limited working proficiency in Spanish

Proficient Technologies:

C, C++, C#, Java, Python, Lua, JavaScript, Rust, HTML, CSS, SQL, Arduino, React, Excel, MQL4, Git, Docker, Linux, Shell

EDUCATION

Queens High School for the Sciences

2021 - 2025
Cumulative GPA: 95.0

New York Institute of Technology

2025 - Present
Cumulative GPA: 3.7

PROJECTS

Flappy Bird Neural Network (C)
github.com/Galm007/flappy_bird_AI

- Implementation of a neural network that autonomously learns to play flappy bird though a genetic algorithm, forked from my flappy bird game made with the Raylib library, written in C.

Triangle Slapper (C)
github.com/Galm007/triangle_slapper

- A program that uses a multi-threaded, genetic algorithm that slaps triangles on top of each other to recreate an image, written in C.
- The algorithm spawns random triangles, selects the best ones based on their calculated scores, and then creates the next generation by having different variants of the current best triangles (a technique commonly used to train AI agents). After a certain amount of generations, the best triangle is drawn into the final image and the cycle repeats.

Forex Data Preprocessor (Python)
github.com/Galm007/forex-data-preprocesor

- This program takes in historical price and volume data, calculates values for commonly used forex indicators, compresses the data into a usable range, and then packs up the data into a neat format for an AI model to use.
- I created this program because my friend wanted to train his custom forex trading AI but couldn't find suitable data to train his model on.

Optimized A Major Algorithm In math.js (JavaScript)
github.com/josdejong/mathjs/pull/3251

- I optimized the library's matrix mapping functions to speed them up by over 300% using optimization tricks that I learned from being a systems programmer. These speedups are crucial for the overall efficiency of math.js because these functions are commonly used inside of the library's many other algorithms.

Metaballs Visualization (Rust and C#)
github.com/Galm007/metaballs_macroquad | github.com/Galm007/MetaBalls

- Metaballs visualization implementation written in Rust and C# using the macroquad library and MonoGame framework, respectively, based on Jamie Wong's article:
  https://jamie-wong.com/2014/08/19/metaballs-and-marching-squares/

Automated Forex Trading Algorithms for MT4 (MQL4)
github.com/Galm007/MT4-Expert-Advisors

- These are the Forex trading algorithms that I've developed for use inside the MetaTrader4 trading platform. The algorithms are ordered by the time when I created them. My best performing algorithms are the ones most recently created.